

Graph Autoencoder Anomaly Detection for E-Commerce Application by Contextual Integrating Contrast With Reconstruction and Complementarity

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Abstract—Considering the personalized characteristics and behavior patterns of users, quickly and accurately identifying abnormal users is crucial in the realm of E-commerce applications. However, existing methods fall short in integrating local and global structural information. Meanwhile, contextual information can serve effectively as a supervisory signal for anomaly detection. This paper proposes a Graph Autoencoder Anomaly Detection model, GAAD, that leverages context-integrated contrast and reconstruction complementarity. This model utilizes local information of target nodes and adopts a contrastive learning module based on graph neural networks to capture anomalies in the structure space; simultaneously, a context sub-graph of the global information of the target node is constructed and anomalies in the attribute space are captured through the autoencoder-based reconstruction module. The GAAD model integrates two complementary modules, contrast and reconstruction, to enhance the accuracy of anomaly node detection. The experimental results showcase the superior performance of the proposed model over existing baseline methods across all seven benchmark datasets, thereby highlighting its efficiency in anomaly detection within E-commerce applications. Our code is available at <https://github.com/jinjiewu2000/GAAD>.

Index Terms—Autoencoder, graph neural network, contextual integrating, anomaly detection.

I. INTRODUCTION

THE APPLICATION of anomaly detection in E-commerce activities is of great significance. For example, in E-commerce financial activities, it is necessary to identify fraudulent accounts; and it's crucial to quickly and accurately identify anomalous users [1] in order to warn about or halt abnormal user behavior. Attribute networks are adept at effectively modeling such application scenarios. In attribute networks, nodes with attributes represent real-world entities, and links represent relationships between entities. However,

the attribute network graph not only encompasses complex connection (structural) information and node attribute details but also typically harbors various types of anomalies, making anomaly detection based on attribute networks a challenging task [2], [3].

Currently, considerable research is dedicated to anomaly detection utilizing attribute networks, and numerous solution strategies have emerged. Shallow anomaly detection methods include techniques like AMEN and Radar. However, shallow networks cannot model complex interdependent graph relationships. With the significant progress made by deep neural networks, researchers have also been exploring the use of deep learning techniques for anomaly detection [4] and abnormal behavior analysis [5]. Deep anomaly detection methods, like Xiao et al. [6], achieve anomaly detection in various environments by learning causal relationships and generalizing them to new scenarios. Jin et al. [7] proposed a model for graph anomaly detection that detects anomalies based on the relationship between nodes and their subgraphs. Zhang et al. [8] detects anomalies by comparing nodes and subgraphs. These methods overlook the contextual information of nodes, and anomalies based on this context can disrupt personalized features. Even if the supervised classification method is used, it is not applicable. Therefore, there remains a deficiency in effective anomaly detection methods capable of handling multiple anomaly patterns in complex scenarios [9], [10].

Deep anomaly detection in attribute networks is commonly conducted using the graph neural network model, GNN [11]. GNN encompasses several models, such as graph autoencoders (GAEs) and graph convolutional networks (GCNs), among others. In the field of deep graph representation learning, the DGI [12] method learns the representation of nodes in the graph by maximizing mutual information. Following DGI, Ren et al. [13] proposed heterogeneous DGI (HDGI), which learns node representations in the graph by maximizing local-global mutual information. Graph autoencoders learn latent representations by reconstructing the structural information of graphs [14]. BLoG [15] utilizes local and global regularization for recommended guided graph representation learning. Although GNNs have made significant progress in areas such as biomedicine, material science, and mechanics, and even extended to applications such as social network analysis and drug discovery [16], advances in graph anomaly detection by existing methods [2] rely mainly on

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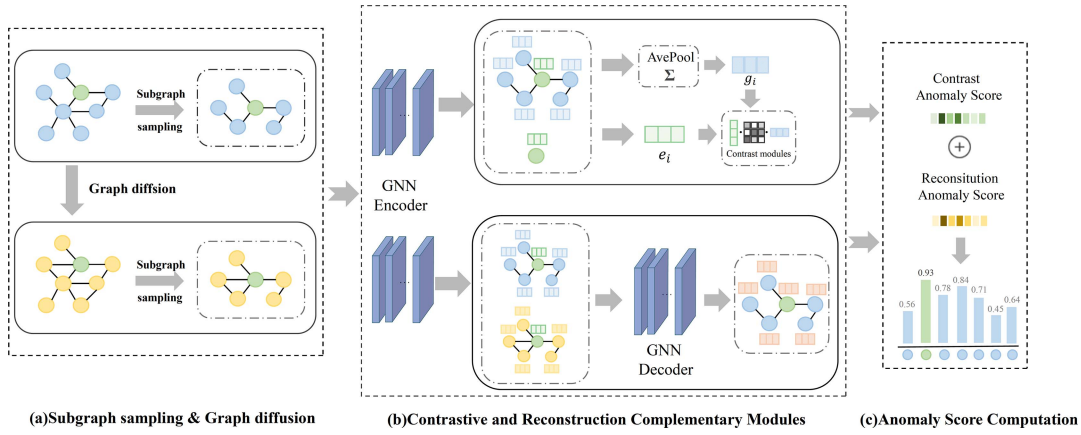


Fig. 1. Overall framework.

purely unsupervised approaches, and still suffer from three shortcomings:

First, local and global contextual information lacks effective utilization and integration. Using instance pairs derived from node subgraph construction and contrast can enhance anomaly detection by effectively attending to local node information. However, this method extracts only the local structural information of the target node, failing to capture and apply its global structure effectively.

Second, existing autoencoder methods based on graph data fail to fully utilize contextual information (e.g., subgraphs or neighboring nodes), which is crucial for anomaly detection. Typically, these autoencoder methods aim to reconstruct attributes or the entire graph structure for each node, disregarding contextual information.

Third, existing methods have inconsistent detection effects on anomalies in the E-commerce field, and there is still room for improvement.

To address these issues, it is imperative to transcend the constraints of current studies, enhance the characterization quality of network structural features, and adopt a more effective, decoupled approach for extracting and integrating local and global structural information. Consequently, this study introduces a model, termed GAAD, that constructs the local and global contexts of target nodes by modeling their respective information and utilizes a self-supervised learning strategy to allow the complementary modules of contrast learning and attribute reconstruction to fully discover and utilize the features and structures in the data to compute each node's anomaly score. To verify the method's efficacy, this study conducts thorough experiments on seven datasets. The workflow of the model is illustrated in Figure 1.

The method described above establishes a new self-supervised graph anomaly detection model by integrating two complementary modules, distinct from existing methods. The primary contributions of this study are as follows:

- 1) A new graph anomaly detection model is proposed to fully utilize the context information of graph data and achieve a more comprehensive capture of anomaly patterns by modeling local and global context information separately.

- 2) A self-supervised learning method is designed to solve the anomaly detection problem in attribute networks. This method can effectively utilize context integration to integrate methods based on attribute reconstruction and contrastive learning.
- 3) Extensive experiments are conducted on seven real-life datasets to demonstrate the effectiveness of GAAD. Experimental results show that the performance of this GAAD is better than the current algorithm, and its anomaly detection effect is superior in the field of E-commerce.

The rest of the paper is structured as follows: Section II describes the problem definition; Section III introduces the proposed anomaly detection model in detail; Section IV discusses the experimental results; and Section V concludes with a summary.

II. PROBLEM DEFINITION

In this article, lowercase letters represent vectors and uppercase letters represent matrices. The attribute network is defined as described below:

Definition 1 (Attribute Network): Given a graph $G=(X, A)$ to represent an attribute network. In this paper, $X \in R^{n \times d}$ represents the attribute matrix of the attribute network, where $V = (v_1, v_2, \dots, v_n)$ is the set of nodes ($|V| = n$), d represents the feature dimension of the attribute matrix X , $A \in R^{n \times n}$ represents the adjacency matrix of the attribute network if there is a link between node i and node j , then $A_{ij} = 1$, otherwise $A_{ij} = 0$.

Definition 2 (Anomaly Detection of Attribute Networks): Given a graph $G=(X, A)$, the goal is to learn an anomaly scoring function $score(-)$, computing the score that the anomaly for each node within G . The anomaly score characterizes the degree of anomaly of a node. By ordering the anomaly scores of all nodes in the graph, anomalous nodes can be detected based on their location.

III. METHODOLOGY

A. Overall Framework

The overall framework of the GAAD method proposed in this article is shown in Figure 1. The framework comprises

three components: graph view sampling, the complementary modules of contrastive learning and attribute reconstruction, and anomaly score computation. The first component, graph view sampling, employs subgraph sampling and graph diffusion to capture local and global view subgraphs of the target node. The second component constructs the complementary modules of contrastive learning and attribute reconstruction to fully exploit node and subgraph contextual information for anomaly detection. Within this component, the attribute reconstruction module utilizes the target node's local and global contextual information to reconstruct its feature vector. If the selected node is anomalous, its features will not align with those of the surrounding environment, resulting in a significant disparity between the reconstructed and original feature vectors. The contrastive learning module directly compares the target node with its surrounding context to ascertain any incorrect links with neighboring nodes. This module captures anomalies within the structural space. The third component integrates the two complementary modules to compute the anomaly score through contrast and reconstruction methodologies for reliable anomaly detection in E-commerce applications. The subsequent sections of this chapter, from Sections III-B–III-E, will delve into the three core components of GAAD, with Section III-F outlining the algorithm.

B. Generation of Multiple Views

This study introduces two self-supervised learning objectives to leverage the target node's extensive global contextual information and form significant connections with its contextual surroundings, with each objective stemming from distinct scales and dimensions. Initially, node-level identification employs GAE to reconstruct the target node's feature representation, which is then contrasted with the actual values present within the attribute network. Throughout the construction phase, the sampling of multiple perspectives allows the contrast module to investigate a variety of contextual data during recognition. The general view is the network structure obtained by reading from the dataset.

As depicted in Figure 1's leftmost section, select the target node in the initial view, select the pair of nodes and their adjacent edges and nodes to form the local view subgraph, and then use graph diffusion to obtain the global view. In the global Select the target node in the view, select the pair of nodes and their adjacent edges and nodes to form the global view subgraph. In the generation target, the identification objects are the original and reconstructed target nodes. Next, we will introduce the method of generating two views:

1) Graph diffusion. The method involves graph diffusion, employing PageRank (PPR), to capture global structural information and obtain a global view.

2) Subgraph sampling. Random walk with restart (RWR) is used to sample the view to obtain the view subgraph, which ensures the preservation of the semantic information of the underlying graph data.

C. Attribute Reconstruction Based on Local and Global Graphs

In the context of graph data anomaly detection, traditional deep autoencoders (DAE) can only reconstruct the attribute

information of nodes, but cannot consider the underlying topology of the nodes, which is not conducive to anomaly detection of graph nodes. To overcome this constraint, as depicted in the central portion of Figure 1, this paper designs a graph autoencoder (GAE) consisting of two components: an encoder and a decoder based on a graph neural network (GNN).

GNN-based encoder. Given a graph view G_i of a target node v_i , use a GNN encoder to convert its high-dimensional node features into a low-dimensional representation, the formula is:

$$H_i = GNN_{enc}(X_i, A_i) \quad (1)$$

where A_i , X_i , and H_i Respectively represent the adjacency matrix, feature matrix, and embedding matrix of G_i . GNN_{enc} is a graphical encoder consisting of multiple layers of GNNs.

To improve the efficiency and avoid the overfitting problem, this paper adopts a single-layer GCN with random weights as the main graph encoder. Therefore, Equation (1) can be specifically expressed as:

$$H_i = GNN_{enc}(X_i, A_i) = \sigma(X_i, \hat{A}_i, W_1) \quad (2)$$

where $\hat{A}_i = (D)^{-\frac{1}{2}} \tilde{A}_i (D)^{-\frac{1}{2}}$, $\tilde{A}_i = A_i + I$. Specifically, $\sigma(\cdot)$ represents a nonlinear activation function, and W_1 is the matrix of trainable parameters in the graph encoder.

GNN-based decoder. In the same way, a single-layer GCN with random weights is used to construct the graph data decoder. During the attribute reconstruction process, this paper takes the node embedding matrix as input to reconstruct the node features:

$$X_i^r = GNN_{dec}(H_i, A_i) = \sigma(X_i, \hat{A}_i, W_2) \quad (3)$$

where W_2 is the matrix of trainable parameters in the graph decoder, GNN_{dec} has the same structure as the GNN_{enc} and X_i^r represents the reconstructed node feature matrix.

Attribute reconstruction anomaly detection. The attribute reconstruction of view G_i is mainly based on the attribute information of the local and global subgraphs centered on node v_i . To achieve attribute reconstruction based on contextual information, this paper anonymizes the graph view of the target node v_i . As mentioned in this paper, this module can better capture node-level exceptions in the attribute space. This paper detects anomalies by minimizing the mean square error (MSE) between the original and reconstructed features of the target node in the graph view, as shown in Figure 1.

$$L_{rec} = \frac{1}{N} \sum_{i=1}^N (x_i^r[-1, :], x_i)^2 \quad (4)$$

where $x_i^r[-1, :]$ is the reconstructed feature vector for the anonymous target node v_i , and x_i is the feature vector of the target node v_i , where x_i^r can be obtained from Equation (2) and Equation (3).

D. Contrastive Learning Based on Autoencoders

To fully exploit the structural information of the nodes, this paper proposes a contrastive learning module that consists of three different components, and uses an encoder and a readout module to extract the low-dimensional embedding of the target

node and local subgraph and then compares the consistency of the target node with its surrounding environment. This module distinguishes the representation of target nodes and their local subgraph representations within the embedding and structure spaces, thereby placing greater emphasis on the surrounding contextual information. The components of the module are as follows:

GNN-based encoder. The contrastive learning module in this paper selects the node feature vector and matrix of the target node v_i , along with the relevant graph view, as inputs. The encoding of the graph view is depicted as in Equation (2), whereas a different formula is employed for the transformation of the target node's feature vector:

$$e_i = \sigma(x_i W_2) \quad (5)$$

where e_i is the embedding vectors of the target node v_i , and x_i is the feature vectors of the target node v_i . Due to the absence of a lower-level graph structure for individual target nodes, the graph adjacency matrix from Equation (3) is omitted in Equation (5), and W_2 is the same as in Equation (3).

Readout module. The goal of this module is to allow embedding vectors e_i to be directly compared to subgraphs, with a variety of readout functions, such as average and micropooling. This paper uses average pooling, which can be expressed as:

$$g_i = \frac{1}{M} \sum_{j=1}^M H_i[j, :] \quad (6)$$

where g_i represents the graph-level embedding vector of the view G_i , $H_i[j, :]$ represents The embedding vector of v_j in H_i , M is the number of nodes in the graph view.

Contrast modules. This module is the central element of the contrastive learning module. It can obtain the degree of difference between two vectors by comparing their embeddings in an instance pair. For contrast e_i and g_i , a contrast module based on bilinear transformation is designed, where the contrast score is defined as follows:

$$S_i = \sigma(e_i W_3 g_i^T) \quad (7)$$

where W_3 is a learnable scoring matrix. A sigmoid function is used as a nonlinear transformation $\sigma(\cdot)$ of the above equation to ensure that the discrimination score is within the range $[0, 1]$.

Anomaly detection with contrastive learning. Detection aims to minimize the difference between the predicted S_i and the true label y_i . Therefore, in this section, the binary cross-entropy loss with Logits is used as the objective function, which combines the Logits activation of the Sigmoid function and the binary cross-entropy loss and can be a good measure of the difference between the model's prediction and the real label.

$$L_{con} = -\frac{1}{N} \sum_{i=1}^N y_i \log(S_i) + (1 - y_i) \log(1 - S_i) \quad (8)$$

E. Anomaly Score Computation

In the module of attribute reconstruction, the reconstruction of the target node's attributes relies on both its local

(i.e., neighboring) and global contextual information, where the degree of disparity between the reconstructed feature vector and the original one serves as a metric for detecting anomalies in the node. This paper uses L2-norm distance as the scoring function for reconstructed anomalies. For node $v_i \in V$, the following function is defined:

$$f_{rec}(v_i) = \gamma \left(\|x_i^r[-1, :] - x_i\|_2^2 \right) \quad (9)$$

Among them, $f_{rec}(v_i)$ is the attribute reconstruction anomaly score of v_i . The score is proportional to the degree of node anomaly. In the above equation, $x_i^r[-1, :]$ is the reconstructed eigenvector of v_i , provided by Equation (3), where γ is a scalar, to ultimately limit the attribute reconstruction anomaly score to the range of $[0, 1]$.

The attribute reconstruction module in this paper only focuses on node discrimination in the attribute space without directly utilizing the topological information of the graph. To overcome this constraint, the contrastive learning module directly distinguishes between the target node and the neighboring subgraphs within the embedding space, with the contrast scores from Equation (7) naturally merging into a unified score one. For node $v_i \in V$, the contrast anomaly scoring function is defined as:

$$f_{con}(v_i) = \delta(\tilde{S}_i - S_i) \quad (10)$$

among $f_{con}(v_i)$ is the contrast abnormality score of v_i , where S_i and $\tilde{S}_i$ are the positive and negative contrast discrimination scores respectively. If the node v_i is normal, then S_i should be close to 1, while $\tilde{S}_i$ is likely to be close to 0. Otherwise, if node v_i is anomalous, S_i and $\tilde{S}_i$ should be close to 0.5. Therefore, the range of $\tilde{S}_i - S_i$ is $[-1, 0]$, but this paper introduces a scalar δ to ultimately limit the contrast learning anomaly score to the range of $[0, 1]$.

Finally, the above two functions are combined to obtain the final anomaly scoring function for the graph data to estimate a node $v_i \in V$ of the anomaly:

$$f(v_i) = \alpha f_{con}(v_i) + \beta f_{rec}(v_i) \quad (11)$$

Among them, α and β are two balancing factors to weigh the importance of the contrastive learning module and the attribute reconstruction module. In practice, this calculation is repeated R times to obtain statistically stable anomaly scores and finally the final anomaly score function.

$$score(v_i) = \frac{1}{R} f(v_i) \quad (12)$$

F. GAAD Algorithm

Combining Equations (5) and (9), the final optimization goal of this paper is:

$$L = \alpha L_{con} + \beta L_{rec} \quad (13)$$

where L is the final training loss. α and β are two balancing factors identical to those in Equation (11).

The complete procedure of the proposed GAAD algorithm is outlined in Algorithm 1. The time complexity of the algorithm is $O(Rnc(\lambda + c))$, where c represents the number of nodes on which the graph view is based, and λ represents the average node degree in the graph.

Algorithm 1 The Proposed GAAD Algorithm

Input: Attribute graph G ; Training rounds T ; Subgraph size S ; Test round R .

Output: Anomaly score function of the graph: $score(\cdot)$

```

1: Randomly initialize trainable parameters  $W_1, W_2, W_3$ ;
2: for  $t \in 1, 2, \dots, T$  do
3:    $S \leftarrow$  Randomly divide  $V$  into batches of size  $S$ ;
4:   for batch  $s = (v_1, \dots, v_B) \in S$  do
5:     Sample local view subgraph and global view subgraphs for each node in  $s$ ;
6:     Calculate node and its associated graph view embeddings via Eq. (2) and (5);
7:     Calculate the reconstructed feature vector of the node in  $s$  via Eq. (3);
8:     Calculate the loss  $L$  via Eq. (4), (8), and (13);
9:     Back propagate and update the trainable parameters  $W_1, W_2, W_3$ ;
10:  end for
11: end for
12: for  $v_i \in V$  do
13:   for  $r \in 1, 2, \dots, R$  do
14:     Calculate  $f(v_i)$  for each round through Eq. (9), (10), and (11);
15:   end for
16:   The final anomaly  $score(v_i)$  is calculated by Eq. (12);
17: end for
```

TABLE I
INFORMATION OF THE DATASET. THE FIRST THREE DATASETS ARE CITATION NETWORK DATASETS, THE LAST THREE ARE SOCIAL NETWORK DATASETS, AND THE LAST ONE IS AN E-COMMERCE DATASET

Dataset	Node	Edges	Attributes	Abnormal
Citeseer	3327	4732	3703	150
Cora	2708	5429	1433	150
ACM	16484	71890	8337	600
BlogCatalog	5196	171743	8189	300
Flickr	7575	239738	12407	450
Twitter	4865	66772	820	300
Amazon	7650	119081	745	600

IV. EXPERIMENTAL STUDY

In this section, this paper assesses the effectiveness of the GAAD model proposed in this study through extensive experiments conducted on seven real-world benchmark datasets. First, the datasets and experimental setups used for the experiments are presented, second, a performance comparison was conducted by evaluating our model against state-of-the-art anomaly detection methods, and finally, ablation experiments and parameter analyses are also performed in this paper.

A. Dataset

To evaluate the performance of GAAD and its comparison algorithm, seven real-world graph datasets are used as benchmarks, including three social network datasets, three citation network datasets, and one E-commerce dataset. A description of these datasets is as follows:

- 1) Citation Network. CORA, Citeseer, and ACM are three citation datasets, which consist of scientific publications.
- 2) Social Network. BlogCatalog, Flickr, and Twitter are three social network datasets, and their data come from three social network platforms respectively.
- 3) E-commerce. Amazon is a product sales dataset, which comes from Amazon, the largest online E-commerce company in the United States. In this dataset, the nodes are products, and the edges represent whether the two products are purchased together. The characteristics of each item are processed as node characteristics.

Since the above dataset does not contain ground truth anomalies, the synthetic anomalies need to be injected into the clean attribute network for evaluation. Table II also summarizes the total number of anomalies as well as the dataset statistics. The dataset is composed of nodes and their connected edges, and its context information is the sampled target node and its adjacent edges and nodes.

B. Experimental Setup

In this section, this paper will describe the experimental setup, including the baseline methods for comparison, evaluation metrics, and fundamental parameter configurations.

This paper compares the proposed GAAD model with seven different anomaly detection methods. The paper briefly introduces these methods as follows:

AMEN [17] utilizes information from the target node with its self-network to detect anomalies. Radar [18] detects anomalies by assessing the consistency between residual attributes and graphical information. ANOMALOUS [19] detects anomalies by combining CUR decomposition and residual analysis. DOMINANT [2] utilizes a graph autoencoder to combine the reconstructed adjacency matrix and the attribute matrix, and calculates the reconstruction error of nodes to detect anomalies. DGI [12] uses a bilinear discriminator to score node anomalies by comparing nodes and subgraphs. COLA [3] detects anomalies by using an encoder model of GNN to learn node embeddings and determine the consistency between the embedding and its neighboring subgraphs. Sub-CR [8] detects anomalies by learning embedded representations of subgraphs and target nodes, followed by multi-view contrast learning, and finally by attribute reconstruction errors in both views.

Evaluation metrics. This paper evaluates the anomaly detection performance of GAAD and its competitors using a common benchmark AUC metric. AUC stands for the area under the ROC curve, with values ranging from 0 to 1. A higher AUC value indicates better detection performance.

Basic parameter settings. To improve performance while ensuring efficiency, this paper sets the size of the subgraph to 4, and a single-layer GCN model is employed as the GNN encoder, with an embedding dimension set to 64. During the training process, for the dataset, α is fixed at 0.8 or 1, then β values are selected from {0.2, 0.4, 0.6, 0.8, 1}, and the ADAM optimizer is used for model training. In terms of learning rate, it is set according to the characteristics of the dataset. Specifically, Cora, CiteSeer, and Flickr use 0.001,

TABLE II
THE AUC VALUES COMPARISON ON SEVEN BENCHMARK DATASETS

method	Cora	Citeseer	ACM	Flickr	BlogCatalog	Twitter	Amazon
AMEN[17]	0.6266	0.6154	0.5626	0.6573	0.6392	0.6265	0.6754
Rader[18]	0.6587	0.6709	0.7427	0.7399	0.7401	0.7384	0.7268
ANOMALOUS[19]	0.5770	0.6307	0.7038	0.7434	0.7237	0.7249	0.7586
DOMINANT[2]	0.8155	0.8251	0.7601	0.7442	0.7468	0.7536	0.7594
DGI[12]	0.7511	0.8293	0.6240	0.6237	0.5827	0.5947	0.6489
COLA[3]	0.8779	0.8968	<u>0.8237</u>	0.7513	0.7854	<u>0.8031</u>	<u>0.8763</u>
Sub-CR[8]	<u>0.9132</u>	<u>0.9303</u>	0.7872	<u>0.7975</u>	<u>0.8141</u>	0.7254	0.7442
GAAD[our]	0.9163	0.9341	0.8292	0.8015	0.8198	0.8206	0.8924

TABLE III
EFFECT OF DIFFERENT READOUT FUNCTIONS ON AUC VALUE

method	Cora	Citeseer	ACM	Flickr	BlogCatalog	Twitter	Amazon
Min Pooling	0.9007	0.8671	0.8116	0.7830	0.7921	0.7997	0.8326
Max Pooling	0.9040	0.8933	0.8114	0.7927	0.8084	0.8140	0.8519
Weighted Average Pooling	0.8927	<u>0.8975</u>	<u>0.8284</u>	<u>0.8003</u>	<u>0.8123</u>	<u>0.8145</u>	<u>0.8888</u>
GAAD(Average Pooling)	0.9163	0.9341	0.8292	0.8015	0.8198	0.8206	0.8924

TABLE IV
EFFECT OF DIFFERENT COMPONENTS ON AUC VALUES. SPECIFICALLY, GAAD-CON, GAAD-REC, AND GAAD-UNWEIGHTED ARE RESPECTIVELY USED TO INDICATE THAT GAAD HAS NO ATTRIBUTE RECONSTRUCTION MODULE, CONTRASTIVE LEARNING MODULE, AND ANOMALY SCORE WEIGHTING

method	Cora	Citeseer	ACM	Flickr	BlogCatalog	Twitter	Amazon
GAAD-CON	0.9128	0.8819	<u>0.8077</u>	0.7570	0.7791	0.7818	<u>0.8675</u>
GAAD-REC	0.8179	0.7856	0.7386	0.7442	0.7464	0.7441	0.7217
GAAD-Unweighted	0.9077	0.8998	0.7977	0.7902	0.8095	0.8003	0.8409
GAAD	0.9163	0.9341	0.8292	0.8015	0.8198	0.8206	0.8924

BlogCatalog, Twitter and Amazon use 0.003, while ACM uses 0.0005. For different datasets, 400 Epochs (BlogCatalog, Flickr, Twitter, and ACM) and 100 Epochs (Cora, CiteSeer, and Amazon) were trained. During the test phase, 256 is used as the sampling parameter R to obtain an accurate evaluation of the model performance.

C. Results and Analysis

The GAAD proposed in this paper is compared with the seven baseline methods in this subsection. The results for AUC values are presented in Table II. The best-performing method in each dataset is indicated in bold and the second best score is underlined. Based on these results, this paper draws the following conclusions:

In all datasets, the GAAD approach presented in this paper surpasses the performance of all baseline methods. In the seven datasets, the average AUC value of GAAD is 0.2375 and 0.1498 higher than the shallow methods AMEN and Radar respectively. The average AUC value in the seven datasets is higher than that of AMEN and Radar. It is 0.051 higher than the best deep detection method in the baseline, demonstrating its ability to detect anomalies with high-dimensional node features and complex structure graphs and its superior detection effect in the e-commerce field. The main reason is that GAAD captures abnormal patterns by jointly using self-supervised methods of reconstruction learning and contrastive learning.

The method proposed in this article demonstrates excellent detection performance and generalization ability. This achievement is partly due to the GAAD method combining two distinct self-supervised learning strategies, which work together to achieve higher learning efficiency. The joint application of this strategy can provide more comprehensive information and help the model more effectively capture abnormal patterns in the data, thereby significantly improving detection performance.

This article repeatedly calculates the abnormality score of each node multiple times to ensure that the final abnormality score is statistically stable and finally detects abnormal nodes based on the abnormality score. In summary, GAAD's comprehensive performance and generalization capabilities make it a compelling innovative work in the field of graph data anomaly detection.

D. Ablation Experiment

In this subsection, the paper studies the impact of varying different readout functions and the impact of various components of the proposed approach.

- 1) In the experiment, this paper examines the impact on the model by altering the selection of various types of readout functions. The readout functions here are min pooling, max pooling, and weighted average pooling. The experimental results are shown in Table IV. The average pooling used in this paper has the best readout

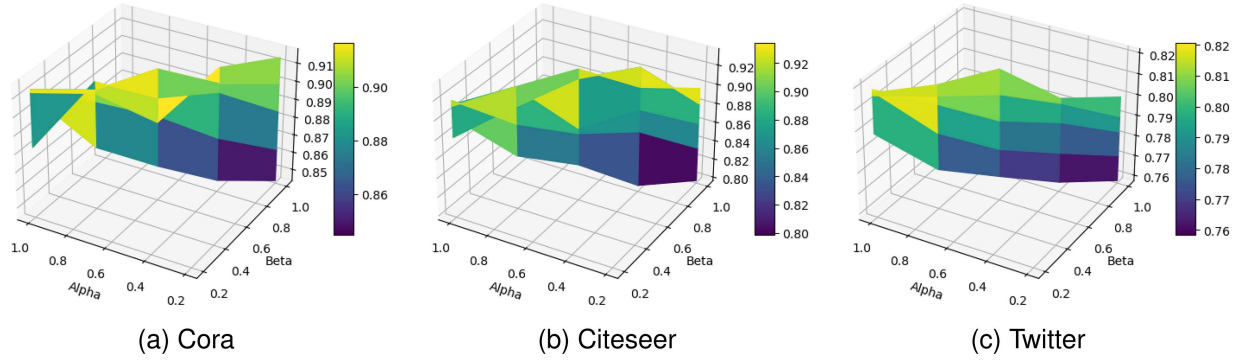


Fig. 2. AUC values of GAAD with different weights of Alpha and Beta on Cora, Citeseer, and Twitter.

effect on the seven datasets compared with other readout functions.

- 2) This paper considers the influence of different components in the model it designs. First consider two variants of GAAD, where GAAD-REC only uses the attribute reconstruction module to identify abnormal nodes and GAAD-CON exclusively employs a contrastive learning module for the identification of anomalous nodes. Finally, to evaluate the effectiveness of the designed anomaly scoring function, this paper further constructs GAAD-Unweighted, which means to obtain attribute reconstruction and comparative learning anomaly scores in an unweighted manner, that is, $\alpha = \beta = 1$ in Equation (13). The experimental results are shown in Table IV and the following points were observed: First, the complete GAAD algorithm achieves the best performance, demonstrating that the two learning strategies in this paper's method are complementary and effective. Second, the weighting of the anomaly scoring function is also helpful in detecting anomalies on the seven datasets. Removing fractional weights results in poor anomaly detection performance. The possible reason is that different components play different effects in detecting anomalies, and $\alpha = \beta = 1$ will lead to poor final detection results. Through ablation experiments, we can first conclude that the readout function used in this article has the best readout effect and has better generalization ability, which is beneficial to subsequent anomaly detection work. Secondly, the effectiveness of the combined method of attribute reconstruction and contrastive learning used in this article in graph anomaly detection is obtained. The combination of these two learning strategies can better identify abnormal patterns in the graph, and finally obtain the effectiveness of the designed anomaly scoring function.

E. Parametric Study

In this subsection, this paper studies the impact of four main parameters on the performance of the GAAD model: the algorithm's balancing factors α and β , test rounds, subgraph size, and the embedding dimension.

1) *Algorithm Balance Factor*: In this experiment, this paper explores how the adjustment of balancing factor α

and β influences anomaly detection performance. This paper separately adjusts these two factors within the range of $\{0.2, 0.4, 0.6, 0.8, 1.0\}$, as shown in Figure 2. The results on the selected dataset are shown here. The results can be obtained to show that the AUC value tends to increase with α when β is constant, and the to reach its peak AUC varies for different datasets. Furthermore, the choice of β is highly dependent on the specific dataset. For example, Citeseer requires a larger value ($\beta \geq 0.6$), while Twitter requires a smaller value ($\beta \leq 0.6$). Depending on the nature of the dataset and the results show that, in practice, this article sets $\alpha = 0.8$ for the Cora dataset and BlogCatalog dataset, $\alpha = 1.0$ for other datasets, and sets the value of β to 0.6 for each dataset.

2) *The Impact of Test Rounds*: In this experiment, this paper studies its effectiveness in the inference phase of GAAD by modifying the test rounds. The value of R is chosen from the set $= \{4, 8, 16, 32, 64, 128, 256, 400\}$. The results can be observed in Figure 3(a). As depicted in the figure, it is evident that inadequate rounds of evaluation result in subpar detection performance. As the number of test rounds increases, the AUC for all datasets steadily increases, indicating the need for an adequate number of rounds to mitigate bias resulting from random sampling. However, when it is greater than 256, there is no obvious increasing trend in AUC. Therefore, maintaining the evaluation rounds at 256 for stable performance.

3) *The Impact of Subgraph Size Factors*: This paper then further explores the impact of subgraph size on AUC. GAAD was run on seven datasets to test the anomaly detection performance under the condition of subgraph size $= \{1, 2, 3, 4, 5, 6, 7, 8\}$. The results are reported in Figure 3(b). From the figure, it can be seen from the figure that when the subgraph size is small, the performance of GAAD sharply increases with the increase of K and quickly reaches the peak. However, when the subgraph size > 5 , the performance decreases as the subgraph size increases. The results indicate that ensuring dependable detection performance hinges on the critical decision of subgraph size. When the subgraph size is small, it is difficult for the model to obtain enough neighborhood information to detect anomalies, while when the subgraph size is large, the subgraph samples unarticulated points around the target node into the subgraph causing it to contain redundant information and affecting the performance.

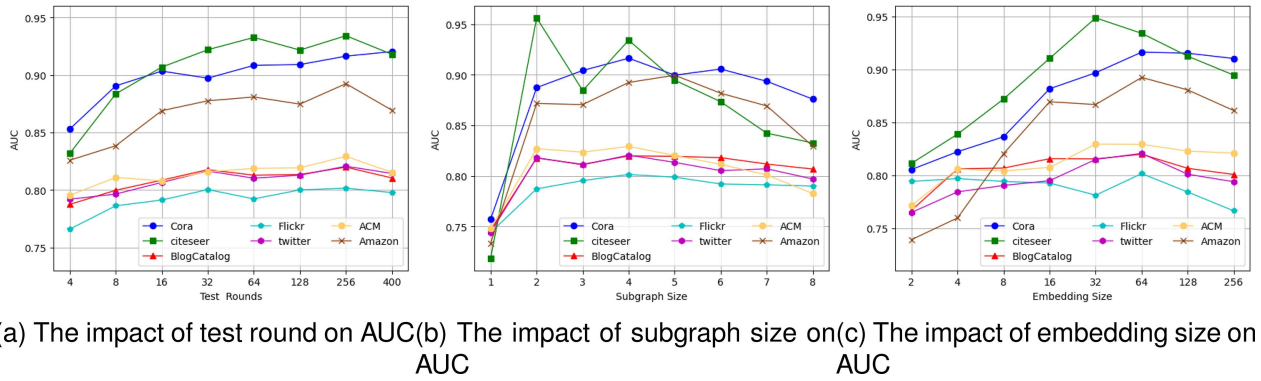


Fig. 3. Parameter sensitivity analysis of GAAD on seven benchmark datasets.

4) *The Impact of Embedding Dimensions:* The paper concludes by exploring the impact of embedding dimensions on the encoder. How it affects the performance of the method in this paper is obtained by varying the value of the embedding dimension, and the results are shown in Fig. 3(c). For most datasets, when $D \leq 32$, performance improves as the embedding dimension increases. When D is larger, the AUC of each dataset has a peak, Then as the embedding dimension increases, the performance starts to decrease slightly. So for most datasets, choosing 64 as the potential embedding is best.

V. CONCLUSION

This paper introduces a novel self-supervised graph auto-encoder learning model, GAAD, for graph anomaly detection, generating global and local subgraphs of target nodes to more effectively capture the structural patterns associated with anomalies. It processes local and global contextual information and then models the structure and attributes of nodes to obtain two key components, the attribute reconstruction module identifies and reconstructs nodes with significant discrepancies, capturing anomalies in the attribute space, while the contrastive learning module detects structural space anomalies. Ultimately, the integration of these two complementary modules seeks to bolster the effectiveness of anomaly detection. Extensive experiments across seven real-world datasets demonstrate that our method can swiftly and accurately detect anomalies, enabling efficient anomaly detection in E-commerce applications. However, the method is unable to detect complex networks such as heterogeneous graphs and dynamic graphs, etc. In the future, we will improve the method to allow it to be applied to more complex network/graph data.

CREDIT AUTHORSHIP CONTRIBUTION STATEMENT

Jinjie Wu: Conceptualization, Methodology, Data curation, Investigation, Experiment, Writing-original draft. **Zhipeng Qiu:** Methodology, Writing-review & editing. **Zhixia Zeng:** data collection, Writing-review & editing. **Ruliang Xiao:** Conceptualization, Methodology, Resources, Writing-review & editing, Project administration. **Imad Rida:** Methodology, Investigation, Writing-Review. **Shi Zhang:** Writing-review & editing.

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